

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination

University Theory Examination January 2018

BPH1002 Pharmaceutical Analysis-I

Date: 05.01.18, Friday

Time: 10:00a.m. to 1:00p.m.

Maximum Marks: 75

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1 Attempt all questions. Each question is of one mark. 20
1. 25 ml of a solution of sodium hydroxide on titration with 0.1 molar solution of hydrochloric acid gave titer value of 35 ml. The molarity of sodium hydroxide solution was
[A] 0.07
[B] 0.14
[C] 0.28
[D] 0.35
 2. The limiting current in a polarogram is proportional to
[A] The reduction potential of the electrode
[B] The standard reduction potential of the solution under investigation
[C] The concentration of the analyte of interest
[D] The standard oxidation reduction potential of the solution under investigation
 3. Molarity of a given orthophosphoric acid solution is 3M. It's normality is
[A] 1N
[B] 2N
[C] 0.3N
[D] 9N
 4. When a pH probe is used during a titration
[A] it is important to stop the titration at the end point
[B] a strong acid and a strong base must be used
[C] an indicator is still necessary
[D] the equivalence point is determined from the shape of the pH curve.

5. _____ titrant used for the assay of sodium benzoate in non-aqueous acid-base titration.

- [A] Perchloric Acid
- [B] Glacial Acetic Acid
- [C] Crystal Violet
- [D] Potassium Permanganate

6. Select the suitable titration method for the quantitative determination of Calcium Gluconate

- [A] Iodometry
- [B] Complexometry
- [C] Precipitation
- [D] Non-Aqueous

7. Gm equivalents of solute dissolved in one litre of solution is called

- [A] Molarity
- [B] Molality
- [C] Normality
- [D] Mole Fraction

8. A solution of sulphuric acid (Mol wt. = 98) consists of 49g of pure acid dissolved to make up a solution of volume 200mL. What is the concentration of this acid solution?

- [A] 0.5mol/litre
- [B] 24.5 mol/litre
- [C] 2.5 mol/litre
- [D] 0.245 mol/litre

9. The pH of a 0.010 M solution of KOH will be

- [A] 13
- [B] 2
- [C] 12
- [D] 0.010

10. EDTA used as a titrant in complexometric titration is classified as

- [A] Unidentate
- [B] Bidentate
- [C] Tridentate
- [D] Polydentate

11. Those substances which are available in pure form with definite chemical composition are called as

- [A] Primary Standard
- [B] Secondary standard
- [C] Pure Compound
- [D] Impure Compound

12. In diazotization titration, the primary aromatic amine reacts with sodium

nitrite to form

- [A] Amine Salt
- [B] Nitrite Salt
- [C] Diazonium Salt
- [D] Azonium Salt

13. The value of pK_w at 25°C is....

- [A] 1.0×10^{-14}
- [B] 1.0×10^{-7}
- [C] 7
- [D] 14

14. The total of oxidation number of atoms or ions in a molecule is

- [A] Zero
- [B] Non-zero
- [C] Volatility of any one atom/ion
- [D] Valency of any one atom/ion

15. Starch Paste is used as _____ indicator in redox titration.

- [A] Self
- [B] Internal
- [C] External
- [D] Self and Internal

16. Errors can be minimised by

- [A] performing parallel determinations
- [B] performing blank determinations
- [C] use of internal standard
- [D] all of the above

17. Which of the following statement(s) is/are true?

- [A] Precipitation is expected to occur if the ion product is greater than the K_{sp}
- [B] The solubility of an ionic compound increases by the addition of a soluble salt that supplies a common ion
- [C] Precipitation is expected to occur if the ion product is smaller than the K_{sp}
- [D] There is no relation between precipitation and K_{sp} value

18. Non-Aqueous titrations are done for

- [A] too strongly acidic or basic substances and substances not soluble in water
- [B] too weakly acidic or basic substances and substances not soluble in water
- [C] too weakly acidic or basic substances and substances soluble in water
- [D] too strongly acidic or basic substances and substances soluble in water

19. _____ is a differentiating solvent for HClO_4 and HCl

- [A] Acetic Acid
 [B] Sodium Methoxide
 [C] Pyridine
 [D] Sodium Hydroxide
20. Which of the following salts dissolves to produce an acidic aqueous solution?
 [A] NH_4Cl
 [B] CH_3OONa
 [C] NH_4OH
 [D] $\text{CH}_3\text{COONH}_4$

SECTION – II

- Q 2 Explain the working of dropping mercury electrode and discuss the different types of currents generated with a suitable polarogram. 10
- Q 3 Attempt any TWO of the following
- A Write a note on factors affecting conductance of a solution. 05
- B Discuss the principle of standardization of sodium hydroxide solution. 05
- C Explain the Standard Hydrogen Electrode with suitable diagram. 05

SECTION – III

- Q 4 Attempt any ONE of the following
- A i) Explain Volhard's method and Fajan's method of precipitation titration and differentiate between the two. 08
 ii) Comment: Nitrobenzene is added in the estimation of chloride by volhard's method. 02
- B Define hydrolysis and derive an equation for finding pH of aqueous solution of salt of weak acid and strong base. 10
- Q 5 Attempt any FIVE of the following
- A Explain in detail Co-precipitation in gravimetric analysis. 05
- B Explain Iodometric and Iodimetric titration (2.5m each) 05
- C Discuss about leveling and differentiating effect of solvents used in non-aqueous titration. 05
- D Explain the principle for assay of calcium gluconate. 05
- E Enlist different types of redox titrations and describe in detail diazotization titration. 05
- F i) Give comment on following (any two) 05
 a. Starch indicator should be added toward the end point in iodometric titration
 b. NH_3 - NH_4Cl buffer is used in the assay of magnesium sulphate
 c. KI is added in preparation of standard solution of iodine.
 ii) Define: Von-Weimarn's ratio and Digestion